

Thesis Executive Summary – Progress Update (23 September 2026)

During the past two weeks, I focused mainly on preparing the technical environment and identifying a suitable dataset for the initial Context Grounding prototype.

I selected Synthea synthetic patient data for the initial experiments and started examining its clinical data structure, including patient demographics, conditions, encounters, observations, procedures, allergies, and medications. The data preparation and preprocessing are still in progress.

I also established a workflow between Google Colab and GitHub and selected Chroma as the initial vector database.

In addition, I created an initial mind map based on the Context Grounding framework and the broader hallucination-reduction directions previously discussed. The mind map is still at an early stage and will be expanded as the literature review progresses.

My academic literature review is still preliminary, as I spent most of this period on the environment, dataset, and initial prototype setup. My next step is to continue the literature review while progressing toward the first working Level 0 Context Grounding prototype.

GitHub Repository: <https://github.com/esrakirbas/msc-thesis-reducing-hallucinations-in-medical-diagnosis>